PROJECT BLUEPRINT / VERSION 1.0

# SakhiLine

## From project brief to verified delivery

A complete phase-by-phase execution guide

12 phases / concrete outputs / acceptance evidence

Voice-assisted documentation for ASHA workers, using invented beneficiaries and synthetic evaluation speech.

**ATHENA / PRODUCT**
Gemini review

**DRACO / SYSTEMS**
Claude review

**HELIOS / DELIVERY**
Codex preparation

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# 1. Executive blueprint

## 1.1 What this plan delivers

This document explains the product, proposed architecture, implementation sequence, ownership split, outputs, acceptance evidence and release conditions from repository initialization to a contest demonstration. Each phase specifies its inputs, tasks, expected result, concrete artifacts, verification and rollback. Desired results are future targets, not accomplished results.

The plan consolidates the nine-page project brief and the three-agent council's frozen R2 recommendation. It adds presentation detail and checklists rather than silently changing approved scope. The original brief remains a vision document; where this MVP reduces that vision, the difference is explicit and requires a human decision.

## 1.2 Current state versus desired state

| Area | Current evidence | Desired result |
|---|---|---|
| Repository | Planning documents; no application manifest or source | Reproducible React/Node TypeScript workspace with tests |
| User workflow | Specified in the brief and council design | Assigned beneficiary to worker-confirmed visit to supervisor approval |
| Speech integration | Public vendor docs read; no authenticated Gnani test in council | Verified short-clip STT and optional deterministic TTS read-back |
| Data/security | Logical model and threat model only | Scoped access, immutable revisions and tested transactional transitions |
| Evaluation | Proposed corpus/protocol; no measured scores | Versioned synthetic artifacts, exact field metrics and honest latency |
| Release | No deployment, public post or submission performed | Human-authorized demonstration with evidence and limitations |

## 1.3 Recommended MVP

- One selected language with English medical terminology where supported; Hindi plus English is a provisional example, not a final commitment.
- Online short-clip capture, transcription and extraction into a tentative form, plus manual correction and text-entry fallback.
- Explicit beneficiary reference confirmation before history is shown; previous readings remain visibly historical.
- Worker review, exact-revision confirmation, idempotent submission, supervisor approve/return and correction/resubmission.
- A small deterministic synthetic robustness evaluation with a transparent results screen.
- Text remains usable when audio playback or TTS fails. Deterministic spoken read-back is included only if verified and within approved budget.

## 1.4 What is deliberately deferred

Continuous live form filling, durable offline audio/text queues, PSTN calls, outbound reminders, custom VAD/barge-in, medical risk flags, clinical thresholds, GPU-hosted Evon, Redis/BullMQ and multiple languages are not MVP dependencies. An online recording demo does not solve no-data capture. A degraded 8kHz audio fixture is not a real telephone integration.

The source brief's live push-to-talk must-ship item is reduced to stop-and-process recording. Human G0 must explicitly accept that change. No feature should be labeled live, offline-ready or clinically validated unless the corresponding behavior was actually demonstrated and tested.

# 2. Product journey and desired experience

## 2.1 Primary users

The worker documents observations after a visit. The supervisor reviews assigned submitted records and returns corrections where needed. A demo viewer inspects reproducible synthetic evaluation evidence. These are product personas, not evidence that interviews, field trials or clinical adoption have occurred.

The initial demonstration is a documentation tool, not a diagnosis, triage, dosage or treatment service. It must not use urgent/emergency-response language or infer care decisions from readings.

## 2.2 Worker journey

1. Sign in through a verified identity integration. The server derives role and assignment; the browser cannot select an authoritative role.
2. Select an assigned invented beneficiary and confirm the synthetic reference/context. Duplicate names require disambiguation; a plausible spoken name is not identity verification.
3. View authorized history labeled by previous visit date. Do not copy old readings into the current visit by default.
4. Record a short clip or enter the approved fields manually. Show record/stop controls, timer, permission state and online-only limitation.
5. Inspect transcript and proposed fields. Resolve units, dates, negation, corrections and current-versus-historical meaning. Preserve manual edits if a provider response arrives late.
6. Review the full snapshot, acknowledge allowed unresolved observations, and explicitly confirm that exact revision.
7. Submit and wait for the durable server receipt containing visit ID, version and pending-review state. A spinner or completed upload is not proof of successful submission.

## 2.3 Supervisor and correction journey

The supervisor sees only assigned pending records. Approval applies to the exact submitted revision, not whichever revision is current after a race. Returning a record requires a reason. The worker reopens it as a new draft revision, corrects it and confirms again before resubmission. Approved records are read-only in MVP; post-approval amendments need a later audited design.

## 2.4 Screen distribution

| Planned route | Output and user purpose |
|---|---|
| /sign-in | Verified sign-in, expired/denied states, safe capability display |
| /visits/new | Assigned roster, identity confirmation, history and capture/manual draft |
| /visits/:visitId | Draft, confirmation, submitted receipt, returned corrections or approved view |
| /review | Paginated assigned supervisor queue, loading/empty/failure states |
| /review/:visitId | Immutable snapshot, approve/return, reason and stale-conflict handling |
| /evaluation | Synthetic run provenance, condition counts, failures, accuracy and latency |

These routes are planned. No route was created during the council or PDF authoring. Each future screen has its own route component and router registration, following CLAUDE.md.

## 2.5 Desired user outcomes and evidence

Provisional prototype targets are at least 90% task completion across at least 10 scripted synthetic tasks, complete keyboard operation and zero observed wrong-beneficiary submissions in that small set. These targets are not results and do not prove clinical safety. Freeze the language, task set and protocol before measuring.

Compare manual entry against voice plus correction on matched invented tasks, including supervisor review time. Report participant count, task count, failed tasks and time distributions. Do not invent a time-saving percentage or claim natural-worker performance from a synthetic evaluation.

# 3. Proposed system and trust boundaries

## 3.1 Architecture at a glance

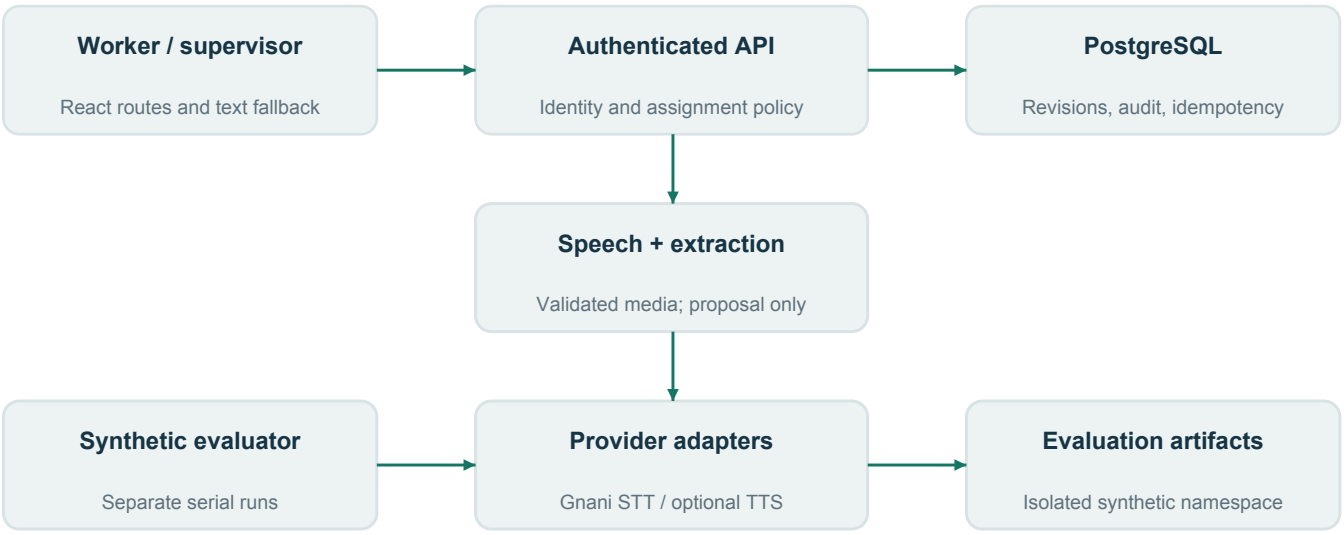



*Figure 1. Proposed application boundaries. Models propose; domain services commit.*

The browser talks to one authenticated Node/TypeScript service. The service performs assignment policy, bounded media handling, provider calls and domain transactions. PostgreSQL holds versioned visits and audit records. The evaluation runner reuses speech/extraction modules but has a separate write namespace and no authority to submit or approve visits.

Proposed stack: React/TypeScript frontend, Express/Node TypeScript backend, shared Zod contracts, PostgreSQL and Prisma ORM. Gnani Prisma STT is a speech model/API and is unrelated to Prisma ORM. Exact dependency versions must be selected and pinned during initialization; none is installed as an application dependency yet.

## 3.2 Non-negotiable boundaries

- A model proposes data; it has no database tools, network tools, role-changing authority or commit authority.
- The server checks verified identity, assignment, schema, semantics, revision and transition for every operation. A hidden button is not access control.
- Submitted content is immutable. Every edit creates a revision and clears confirmation; stale update/review requests fail rather than overwriting newer work.
- Idempotent retries return the same durable result. Reusing a key with a different request body conflicts.
- Historical context is labeled as history. Missing current values stay missing, unknown or not measured rather than inheriting old observations.
- Free model-generated follow-up or advice text is never rendered or spoken. Use approved templates and deterministic read-back from confirmed typed fields, excluding arbitrary notes.

## 3.3 Speech contract and uncertainty

The source verification records the short-clip REST STT contract, documented audio formats and separate raw-PCM streaming protocol. A browser's usual recorder output must not be assumed compatible. T02 verifies actual MIME/codec/duration/channels and selects direct supported capture or bounded conversion. Renaming a WebM file as WAV is not conversion.

Proposed application bounds are 30 seconds and 10MiB per upload, stricter limits taking precedence if required by the provider. A provisional operation timeout of 30 seconds and at most one budget-authorized retry need validation

during the spike. A timeout or cancellation does not prove the provider stopped processing or billing.

Provider keys remain server-only. No paid Gnani/extraction request, key creation, GPU rental or account change is authorized by this document. Public documentation is not an integration test.

## 3.4 Visit lifecycle

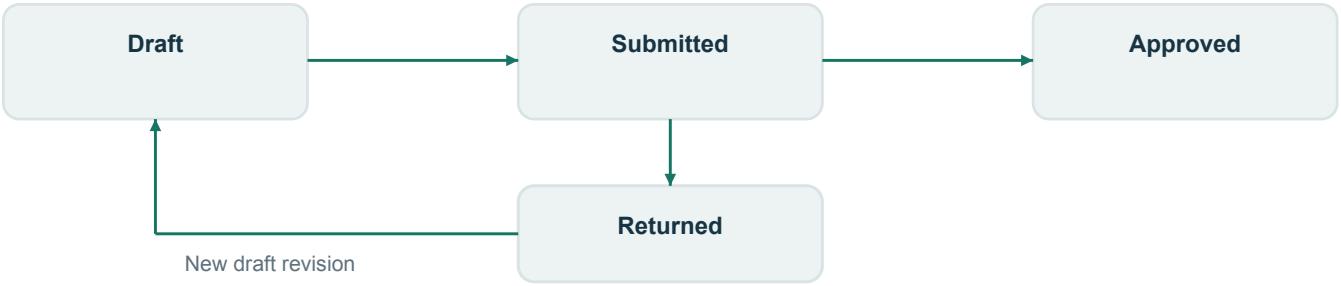



*Figure 2. Version-bound confirmation and explicit return/correction lifecycle.*

Confirmation is a property of an exact revision, not permission for future model updates. Submit atomically checks confirmation, changes state, records audit and persists the idempotent response. Review checks supervisor assignment and the same submitted version. Reopening a returned record creates a new draft. No long-running provider call holds a database transaction open.

# 4. Distribution, estimates and gates

## 4.1 Two builders, three review roles

Athena/Gemini owns product review; Draco/Claude owns systems review; Helios/Codex owns implementation feasibility and integration review. These council roles do not mean three human builders exist.

The source assumes two builders. Proposed distribution: Builder A owns speech/media/extraction and evaluation; Builder B owns UI, data, authorization and review. Both review field semantics and run the end-to-end demonstration. Human availability and skill allocation must be confirmed at G0. Shared contracts and integration checkpoints prevent separate frontend/backend assumptions from drifting.

## 4.2 Phase map

| Phase / task | Main result | Effort in builder-days |
|---|---|---|
| 00 / T00 | Human scope, field-review and budget decisions recorded | 0.5-1 |
| 01 / T01 | Reproducible application workspace and fixture smoke | 0.5-1.5 |
| 02 / T02 | Verified recorder/provider contract and pinned candidate | 0.5-2 |
| 03 / T03 | Typed data, scoped identity and initial synthetic schema | 2-3 |
| 04 / T04 | Accessible worker capture/manual-entry experience | 1.5-3 |
| 05 / T05 | Typed extraction, provenance and safe clarification | 1-2 |
| 06 / T06 | Transactional revision, confirmation and submission | 2-3 |
| 07 / T07 | Supervisor return/correction/approval loop | 1-2 |
| 08 / T08 | Reproducible synthetic evaluation and report | 2-4 |
| 09 / T09 | CI, security, accessibility and device release evidence | 1.5-3 |
| 10 / T10 | Evidence-backed demo, setup and handoff package | 1-2 |
| 11 / T11 | Separately authorized release, post and submission | 0.5-1 |

The task ranges total 14-27.5 builder-days, rounded to approximately 14-28 in the roadmap, excluding approval delays. They are estimates, not measured productivity. Two builders can overlap some tasks but cannot divide all durations by two because shared contracts and review gates create dependencies.

## 4.3 Dependency path and parallel work

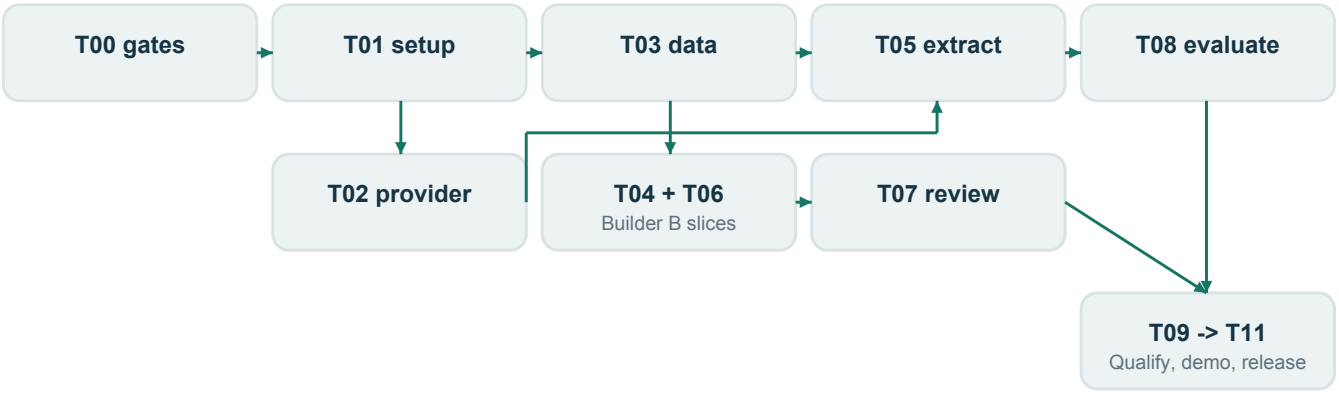



*Figure 3. Key dependencies; shared-builder work must be resource-scheduled.*

T00 decisions and G2 precede any scaffold. After T01, Builder A can run T02 while Builder B implements T03 within approved choices. T05 needs T02 and T03. T04 and T06 depend on T03 and may run in coordinated slices. T07 needs T04 and T06. T08 needs T05 and a frozen schema. T09 follows the integrated review/evaluation paths, then T10 and T11 follow in order.

T04 and T06 are both assigned to Builder B: interleaving small integration slices does not make that builder available for simultaneous full-time tasks. The upper-bound dependency path T00-T01-T03-T05-T08-T09-T10-T11 is 17.5 builder-days before approval delay; resource contention can lengthen elapsed delivery. This is a dependency estimate, not a simulated schedule or promised finish date. Builder A/B workloads and critical path must be rebalanced with actual availability before committing dates.

This is dependency-aware distribution, not a requirement to finish every row serially. Integrate fixture contracts early so a real provider is not necessary to test the worker/review domain. Do not create production code before G2 merely because fixture mode avoids spending.

## 4.4 Human decision gates

| Gate | Human decision | Current status |
|---|---|---|
| G0 | Synthetic contest scope, language, two builders, deadline and recording-first reduction | Pending |
| G1 | Qualified field reviewer, identity/hosting, retention, provider access and bounded spend/retries | Pending |
| G2 | Approve exact plan and explicitly authorize implementation | Closed |
| G3 | Accept executed release checks; separately authorize deployment, ongoing cost, posting/submission | Closed |

Council R2 approval recommends the design and these gates. It does not open any gate. Real data, clinical policy, credentials, destructive migrations or security-policy changes require new explicit review/authorization.

## 4.5 Conditional calendar and cut line

If decisions and authorization arrive promptly, the roadmap reserves October 19-20 for gates/scaffold/spike, October 21-25 for the core loop, October 26-29 for evaluation, October 30-November 2 for hardening, November 3-6 for freeze/demo and November 7-10 for submission buffer. The source deadline is November 10, 2026 at 23:59 IST; recheck current official terms before release.

If approvals slip or the spike fails, reduce optional TTS/report volume or defer nonessential polish openly. Preserve authorization, confirmation/idempotency, privacy, accessibility and honest evidence. Do not add offline, telephony or GPU work late to compensate for a weaker demo. No calendar commitment is valid until actual builder availability is confirmed.

If G0/G1/G2 are unresolved at the end of the planned readiness window, rebase dependent dates from the actual approval time; do not compress verification to preserve the original date labels. Report whether the remaining window still supports the unchanged MVP and escalate a scope/deadline decision if it does not. Builder-day ranges exclude this waiting time.

# 5. Phase-by-phase execution guide

## Phase 00. Scope, readiness and human decisions

Task mapping: T00. Accountable role: Athena with human/domain owner. Estimate: 0.5-1 builder-day plus human response time. Current status: planning discussion only; exit decisions are not yet recorded as approved.

Threat coverage: S-12 authorization/scope growth. Acceptance evidence format: dated written decision/gate register, not inferred consent.

### Purpose and entry conditions

Translate the brief into an agreed deliverable that two actual builders can finish. Inputs are the source PDF, factual repository audit, PRD, risk register and unanimously recommended R2. This phase can proceed as planning; no credential creation, paid tests or application code is needed.

### Work distribution and actions

1. Human chooses contest-only synthetic demo or a different pilot scope. A real-data pilot triggers a new plan, not a relabeling of this one.
2. Select one language and a fluent reviewer. Confirm two builders, available hours and responsibilities.
3. Explicitly accept or reject stop-and-process recording in place of the brief's live form filling. If live is mandatory, re-estimate dependencies and verification.
4. Name the field/unit reviewer and freeze the narrow initial observation set. No clinical thresholds are invented by the engineering team.
5. Decide identity/hosting direction, staging retention and a numerical provider spend ceiling/retry allowance. Establish who can authorize exceptions.
6. Record G0/G1 choices and request explicit G2 implementation authorization for the agreed revision.

### Desired result

Everyone can state the same MVP, exclusions, language, staffing and acceptance evidence. Remaining unknowns are named rather than disguised as defaults. No builder begins unauthorized implementation or paid exploration.

### Concrete outputs

- Updated DECISION_LOG.md recording the human's decisions, date, scope revision and exact authorization boundaries.
- Updated PRD.md/DESIGN.md with chosen language and accepted online limitation.
- Named domain reviewer, builder ownership map, approved spending envelope and selected integration choices.
- Gate checklist showing accepted, rejected and unresolved items; no fabricated signature or inferred consent.

### Exit evidence, failure response and handoff

Exit requires actual written G0/G1 decisions and G2 authorization before T01 starts. If a choice is missing, retain it as blocked and continue only independent planning. Rollback is a documented scope revision, not deleted history. Handoff to Builder A/B includes source revision, decisions and explicit out-of-scope features.

# Phase 01. Initialize the reproducible workspace

Task mapping: T01. Accountable role: Helios. Estimate: 0.5-1.5 builder-days. Dependencies: G0, G1 and G2. Current status: not started.

Threat coverage: S-02 secret exposure, S-12 scope growth and fixture/live identity boundary. Evidence format: clean-install/build/test logs and manifest/lockfile diff.

## Purpose and inputs

Create the smallest complete foundation for a React/TypeScript app, Node/TypeScript service, shared contracts and PostgreSQL access. Input is the authorized plan and selected compatible toolchain. Existing planning files and local October configuration remain intact.

## Work distribution and actions

1. Pin supported Node/npm and dependency versions after checking compatibility. Use npm workspaces and a committed lockfile; no global-only runtime imports.
2. Create apps/web, apps/api, packages/contracts and packages/db manifests with declared dependencies and clear responsibility boundaries.
3. Add typecheck, build and meaningful test scripts. A fixture-mode web-to-API smoke must exercise an actual request/response boundary.
4. Create .env.example with names and harmless comments only; ignore secrets, staging, dumps and generated output while retaining existing ignore rules.
5. Keep fixture and live modes explicit. Remote/live readiness must refuse fixture identity injection and invalid identity configuration.
6. Document the start/stop/reset sequence. Do not scaffold all future channels or add Redis/GPU infrastructure.

## Desired result

A clean checkout can install, compile and run a minimal fixture demonstration without undocumented machine state. Frontend and backend agree on one shared contract. This is setup proof, not working speech or completed visits.

## Concrete outputs and target paths

- Root package.json, package-lock.json and tsconfig.base.json.
- apps/web/package.json, apps/api/package.json, packages/contracts/package.json and packages/db/package.json.
- .env.example and reviewed .gitignore update, plus a first cross-boundary smoke test.
- Updated SETUP_AND_INITIALIZATION.md with actual tested Node version and commands.

## Exit evidence and rollback

Proposed checks after scripts exist: npm ci; npm run typecheck; npm test; npm run build. Report actual output/exit codes and the smoke's asserted behavior. Verify every imported package is declared. Do not call a starter page a complete product. If setup fails, revert the smallest scaffold change while preserving docs/tool configuration. T02 and T03 may proceed in parallel only after this foundation works.

# Phase 02. Verify speech, recorder and extraction contracts

Task mapping: T02. Accountable role: Draco; Builder A executes. Estimate: 0.5-2 builder-days. Dependencies: T01, selected language, actual account access and explicit paid-test budget. Current status: not started.

Threat coverage: S-02 secrets, S-07 media abuse, S-08 billing abuse and S-09 staging leakage. Evidence format: redacted recorder/codec report, bounded conversion timing/size data and authorized smoke metadata.

## Purpose and inputs

Resolve the most consequential integration unknown before a voice experience is built around it. Inputs are approved credentials configured by the authorized owner, synthetic clips, the selected browser/device and public provider documentation. Do not print or commit credential values.

## Work distribution and actions

1. Inspect the actual browser recording container/codec. Validate MIME, decoded format, duration, channels and byte size on the server rather than trusting extension/client declarations.
2. Test a documented supported format directly or a bounded conversion path. Record conversion tooling, CPU/time/resource bounds and cleanup behavior.
3. Make only the authorized tiny synthetic STT request. Capture safe request metadata, response shape, timing and usage delta, never keys or sensitive payload logs.
4. If TTS is in the approved budget, verify one selected voice/model/language and browser playback. A fluent reviewer checks spoken numbers; equivalent visible text remains available.
5. Compare the approved small transcript fixture set on candidate extraction models, or validate one selected model if that is the human choice. Separate valid-JSON rate from correct-field rate.
6. Pin the successful contract, provider/model/prompt/schema identity, error behavior and timeout/retry bound. Do not silently substitute another model.

## Desired result

There is an evidenced recorder-to-provider path and a bounded extraction choice. Unsupported browser media, credit denial or bad pronunciation is discovered early. A failed test remains a failed test; fixture mode may continue but is never represented as live inference.

## Concrete outputs and target paths

- apps/api/src/providers/gnani.ts and apps/api/src/media/ with narrow tested contracts.
- tests/contract/ fixture responses for success, invalid media, malformed output, auth/quota and timeout.
- docs/source/VERIFICATION.md updated with actual authorized test evidence, device/browser version and pinned configuration.
- Provider spike report including actual usage, remaining uncertainty and go/no-go for speech/TTS.

## Exit evidence, failure response and rollback

Run npm run test:contract and an explicitly authorized live smoke, plus the selected device recorder/playback check. Validate failure classification and deletion of staging files. On failure, disable the unverified adapter, retain clearly labeled fixture/manual input and reopen scope if live Gnani use cannot be shown. No new spending or credential rotation occurs without approval. T05 waits for this contract and T03's schema.

# Phase 03. Establish data, identity and assignment policy

Task mapping: T03. Accountable role: Draco; Builder B executes. Estimate: 2-3 builder-days. Dependencies: T01 and G1 field/identity choices. Current status: not started.

Threat coverage: S-01 cross-actor access, S-02 identity secrets, S-05 wrong identity/value, S-06 consistency and S-10 session/web attacks. Evidence format: reviewed migration SQL, seed/constraint logs and actor-denial matrix.

## Purpose and inputs

Create the trusted foundation for every visit and review. Inputs are approved field/unit semantics, chosen identity integration, logical schema and synthetic fixture records. There is no old application data to migrate.

## Work distribution and actions

1. Define shared tagged observations: recorded, not_measured, unknown and not_mentioned. Typed recorded values have declared units and provenance; no default zero/false substitutes missing meaning.
2. Implement verified identity-to-actor mapping and centralized organization/assignment checks. Role comes from server policy, not browser hints.
3. Define Actor, Beneficiary, Assignment, Visit, VisitRevision, TransitionAudit and IdempotencyRecord. Keep EvaluationRun/Result isolated from visit mutation authority.
4. Generate and review the initial additive migration, unique keys, compound organization-consistent FKs and indexes. Use a disposable local database for migration tests.
5. Seed invented actors/beneficiaries/assignments deterministically. Multiple workers and supervisors are necessary to test denials, not just a single happy-path account.
6. Verify remote/live mode cannot accept fixture identity injection. Define session expiry/logout, CSRF/origin and safe error behavior for the chosen identity integration.

## Desired result

Each request has an authenticated, scoped actor and each field preserves the difference between an actual observation and missing information. The schema supports immutable revisions, audit and idempotency before the UI depends on them.

## Concrete outputs and target paths

- packages/contracts/src/visit.ts with semantic test fixtures.
- packages/db/prisma/schema.prisma, migrations/0001_initial/migration.sql and seed.ts.
- apps/api/src/auth/ and apps/api/src/policy/assignments.ts.
- Initial schema/seed migration report and role/assignment-denial matrix.

## Exit evidence and rollback

Proposed checks: unit tests, npm run test:integration, npm run db:migrate:test and repeated synthetic seeding. Prove cross-actor history/detail/write denial, organization consistency, invalid-session rejection and remote fixture-auth rejection. A UUID is not an access token. Roll back application code while keeping compatible additive schema/audit; no destructive production reset/down migration is authorized. T04/T06 can consume the stable contract; T05 also needs T02.

# Phase 04. Build the accessible worker experience

Task mapping: T04. Accountable roles: Athena and Helios; Builder B executes. Estimate: 1.5-3 builder-days. Dependencies: T03. Current status: not started.

Threat coverage: S-01 history disclosure, S-05 wrong identity/value and S-10 untrusted content. Evidence format: E2E trace/screenshots plus manual keyboard, language and device checklist.

## Purpose and inputs

Turn the domain contract into a usable worker journey. Inputs are approved screen design, scoped session/roster/history endpoints and typed fixtures. Use a visible fixture provider until the live integration is ready; do not hide that distinction in a demo.

## Work distribution and actions

1. Create separate sign-in, new-visit and visit routes and register them in the router. Do not build all screens as one internal-state page.
2. Show only assigned synthetic references, disambiguate duplicate names and require reference/context confirmation before fetching history.
3. Add Record/Stop/timer, microphone-denied/silence/unsupported-format/network states and manual entry. State that the first version needs an online connection.
4. Show transcript, previous history and current proposed values as distinct content. Use visible labels for source and unresolved status rather than color alone.
5. Preserve in-memory edits across recoverable errors while the page remains open. Do not imply page-close durability or persistent offline queueing.
6. Implement keyboard navigation, visible focus, accessible labels/status announcements, text alternatives, font zoom and large primary controls. Have a fluent reviewer check language/numerical content.

## Desired result

A worker can navigate a synthetic visit task and understand what is saved, proposed, unknown, confirmed or failed. Speech is an input option, not a barrier to completing the same form by text.

## Concrete outputs and target paths

- apps/web/src/routes/SignIn.tsx, NewVisit.tsx and Visit.tsx, each as a separate file.
- apps/web/src/router.tsx registration and accessible form/capture states.
- E2E fixtures for ambiguous reference, mic denial, network failure and manual correction.
- Device/accessibility checklist with actual observations, not compliance claims.

## Exit evidence and rollback

Run the relevant npm run test:e2e and npm run test:a11y checks after they exist; perform manual keyboard/screen-reader/mobile checks. Later phases may be stubbed through honest fixture contracts, but submission success cannot be mocked as durable live success. Required evidence includes identity-before-history and edit preservation. If capture fails, keep the authorized manual path; disable the faulty feature without losing saved server revisions.

# Phase 05. Extract typed observations safely

Task mapping: T05. Accountable roles: Draco and Helios; Builder A executes. Estimate: 1-2 builder-days. Dependencies: T02 and T03. Current status: not started.

Threat coverage: S-03 prompt injection, S-04 advice-shaped output and S-05 wrong observations. Evidence format: fixed transcript/expected/actual fixture table with separate shape-validity and correct-value counts.

## Purpose and inputs

Convert a frozen transcript into tentative structured observations without giving the model authority over the application. Inputs are the pinned extraction contract, reviewed schema, authorized historical context and fixed visit time/timezone.

## Work distribution and actions

1. Quote transcript/history as data and mark history explicitly old. Never accept embedded instructions to change roles, query other records or submit a visit.
2. Require typed structured output, reject unknown keys and validate semantic meaning as well as JSON shape.
3. Preserve reported numbers, units, negation, corrections and provenance. Relative dates resolve against declared visit context and require clarification when ambiguous.
4. Distinguish not mentioned, unknown and not measured. Never fill missing current readings from prior visits or invented clinical defaults.
5. Return a proposal bound to its base revision. A late result cannot overwrite manual edits; accepting it is an explicit draft action subject to version checks.
6. Generate clarifications from approved fixed templates. Optional read-back uses confirmed typed fields, excluding arbitrary notes and free model text. Keyword filtering is supplemental only.

## Desired result

The model produces inspectable, correctable proposals and unresolved issues rather than an authoritative record. Failure is visible, and the worker can continue manually without a silent model switch.

## Concrete outputs and target paths

- apps/api/src/extraction/ with pinned prompt/model/schema identity.
- apps/api/src/routes/extractions.ts with bounded input and stable failure envelopes.
- tests/unit/extraction.test.ts covering exact-value and ambiguity behavior.
- Typed proposal response, source-span convention and clarification/read-back template catalog.

## Exit evidence and rollback

Unit/contract tests must cover wrong units, number corrections, negation, old/current conflict, prompt injection, malformed JSON, timeout and stale base revision. Schema-valid output is not automatically correct. Disable extraction and preserve manual validated entry if the provider fails; do not turn fixtures into undisclosed live fallback. Handoff to T06 includes proposal shape and edit/confirmation semantics.

# Phase 06. Make confirmation and submission transactional

Task mapping: T06. Accountable role: Helios; Builder B executes. Estimate: 2-3 builder-days. Dependencies: T03; integrate with T04/T05 as ready. Current status: not started.

Systems-review checkpoint: Draco reviews the transactional implementation and real-DB race evidence before T07 relies on it. Helios remains implementation-accountable; this makes D03's joint systems/implementation responsibility explicit without adding another builder.

Threat coverage: S-01 scoped writes, S-05 unconfirmed values and S-06 duplicate/stale transitions. Evidence format: concurrent response logs, before/after row counts and expected 409 conflict assertions.

## Purpose and inputs

Protect the core record from duplicates, stale screens, unconfirmed submission and late inference. Inputs are typed immutable revisions, assignment policy and the approved lifecycle. This phase is the consistency boundary for every future channel.

## Work distribution and actions

1. Route mutations through apps/api/src/domain/visits.ts, which invokes transactional repository functions in packages/db. Only that repository layer holds Prisma ORM access.
2. On edit, verify expected version/state and scope, create a new immutable content revision, advance the pointer, clear confirmation and append audit atomically.
3. On confirmation, bind actor, exact revision/content hash and unresolved acknowledgements. Any subsequent edit invalidates it.
4. On submit, check confirmation and scope; write state transition, audit and original idempotent response in one transaction.
5. Scope idempotency by actor, operation and resource plus request hash. Same key/body returns the original result; changed body conflicts.
6. Reject stale version and invalid transition attempts with stable conflict errors. Do not keep a DB transaction open during STT/extraction calls.

## Desired result

A dropped connection can be retried without creating another logical visit. A worker never submits unseen later model output, and a supervisor cannot approve a changed revision accidentally.

## Concrete outputs and target paths

- apps/api/src/domain/visits.ts, apps/api/src/routes/visits.ts and repository transaction functions.
- tests/integration/visits.test.ts exercising a real disposable database.
- Durable confirmation/submission receipts, audit entries and idempotency records.
- Updated API error/version contract and race-test evidence.

## Exit evidence and rollback

Demonstrate concurrent duplicate submission, same-key/different-body conflict, stale edit, edit-after-confirm invalidation, submission without confirmation and late proposal rejection. Mocks alone do not prove atomic database behavior. Revert application code with compatible schema intact; never delete audit/revisions to simplify rollback. Release is blocked if a race permits duplicate or unconfirmed transitions.

# Phase 07. Complete supervisor review and correction

Task mapping: T07. Accountable roles: Athena and Draco; Builder B executes. Estimate: 1-2 builder-days. Dependencies: T04 and T06. Current status: not started.

Threat coverage: S-01 cross-supervisor access, S-05 unseen snapshot approval and S-06 stale transitions. Evidence format: full correction/resubmission E2E trace and concurrent review/audit assertions.

## Purpose and inputs

Close the complete user task, including correction and resubmission. Inputs are immutable submitted revisions, scoped supervisor assignments, durable receipts and an accessible worker view. This is review of documentation, not clinical triage.

## Work distribution and actions

1. Implement a paginated assigned review queue with loading, empty and error states; no invented clinical-risk ranking.
2. Show the exact submitted snapshot and provenance. Approval/return controls include an expected revision precondition.
3. Require a bounded reason for return. Preserve the record and audit rather than editing a submitted snapshot in place.
4. Reopen a returned visit as a new worker draft, show the return reason and require new confirmation before resubmission.
5. Resolve simultaneous supervisor actions by letting one valid transaction win and giving the stale actor a refresh/conflict response.
6. Keep approved records read-only. Do not implement unaudited post-approval edits under a generic save action.

## Desired result

An invented visit can travel from worker capture to supervisor return, worker correction and final approval with attributable version history. Neither role can access another assignment's records.

## Concrete outputs and target paths

- apps/web/src/routes/ReviewQueue.tsx and ReviewVisit.tsx with separate router entries.
- apps/api/src/routes/reviews.ts and tests/e2e/review.spec.ts.
- Review/return/reopen audit trail and the complete end-to-end correction journey.
- Screenshot or test trace of stale approval and cross-supervisor denial using synthetic fixtures.

## Exit evidence and rollback

E2E and database integration checks prove exact-version approval, return reason, fresh confirmation, cross-supervisor denial and simultaneous review conflict. Include keyboard/manual path checks. If review mutations fail, disable them explicitly and preserve pending submissions; do not auto-approve or discard records. This phase supplies the core live demonstration path to T09/T10.

# Phase 08. Produce reproducible synthetic evidence

Task mapping: T08. Accountable roles: Draco and Helios; Builder A executes. Estimate: 2-4 builder-days. Dependencies: T05, schema freeze and explicit generation/evaluation budget. Current status: not started.

Threat coverage: S-08 evaluation spending and S-11 data/metric contamination. Evidence format: scorer self-test log, frozen manifest hash, versioned run output and denominator report.

## Purpose and inputs

Measure the pipeline honestly under controlled synthesized speech conditions. Inputs are independently reviewed invented expected records, approved scripts, a pinned provider configuration and deterministic degradation recipes. No human/volunteer voice recording is collected.

## Work distribution and actions

1. Freeze 30 base scenarios: 20 development and 10 held out. Split by underlying scenario before paraphrases/noise so a held-out variant cannot leak into tuning.
2. Prepare clean plus two fixed noise conditions for a proposed 90 cases if budget permits. Document sample rate/codec/noise recipe, seed and artifact hash; reduce volume explicitly if the ceiling cannot support it.
3. Author expected JSON before speech generation. The generator never decides whether its own answer is correct. Cache speech to avoid regenerating identical input.
4. Reuse actual STT/extraction modules but isolate evaluation writes from visit/submission authority. Run concurrency 1 separately from interactive demonstration sessions.
5. Score exact structured values after declared normalization. Track correct, wrong, missing, abstained, schema-failed and provider-failed outcomes; do not fuzzy-match a wrong number into a pass.
6. Publish run provenance, per-field denominators, condition counts, coverage and record-stop-to-editable p50/p95. TTS first audio is separate. Label results synthetic robustness/self-consistency, not clinical or natural-worker accuracy.

## Desired result

Every demonstrated accuracy/latency statement traces to a fixed run and artifact set. Failures remain visible in end-to-end denominators. A rerun with the same fixtures and versions can explain differences rather than silently changing the dataset.

## Concrete outputs and target paths

- eval/fixtures/manifest.json with base IDs, split, expected records, time context, conditions/seeds/hashes.
- eval/src/scorer.ts, eval/src/run.ts and eval/tests/scorer.test.ts.
- docs/evaluation/ result artifacts, run metadata and a human-readable methods/limitations report.
- apps/web/src/routes/Evaluation.tsx with actual measured results or explicit no-results state.

## Exit evidence and rollback

npm run eval:self-test must catch swapped numbers, wrong units, negation, missing values, bad denominators and split leakage. A budget-authorized real synthetic run produces the report; fixture-only runs must be labeled. Verify no evaluation operation can create/approve a visit. If versions change, mark old results obsolete for the new build, retain provenance and rerun only within approval. Never replace missing results with attractive sample numbers.

A passing scorer self-test means the measurement logic correctly detects its injected failures. It does not mean STT/extraction accuracy is acceptable. R2 intentionally sets no measured field-accuracy pass bar before real runs. Human G3 must inspect actual results and limitations; any requested numerical accuracy acceptance bar should be agreed before evaluation, not chosen afterward to make results pass.

# Phase 09. Harden, integrate and qualify the build

Task mapping: T09. Accountable role: Helios; both builders execute. Estimate: 1.5-3 builder-days. Dependencies: T07 and T08. Current status: not started.

Threat coverage: S-01 through S-12 integrated verification as applicable, with explicit blocked/not-applicable reasons. Evidence format: CI results, security/secret inspection, fixture-load metrics and cleanup/device/accessibility reports.

## Purpose and inputs

Convert independently working components into an evidenced release candidate. Inputs are the integrated worker/review loop, provider contracts, synthetic corpus, threat model and selected target devices. Tests are applicable requirements, not a promise that every risk has been eliminated.

## Work distribution and actions

1. Add CI clean install, typecheck, unit/contract/integration/E2E build gates and synthetic seed/migration smoke.
2. Exercise cross-role/resource denial, invalid sessions, CSRF/origin, prompt injection, output escaping, oversized/spoofed media, conversion timeout and remote fixture-auth rejection.
3. Check real DB concurrency/idempotency and the correction/approval flow. Preserve failure evidence rather than weakening assertions.
4. Inspect browser bundle and logs for secrets/payload leaks. Logs contain only safe operation/request IDs, versions, timings and error classes.
5. Verify private audio cleanup on success/failure and crash-recovery sweeps. Proposed maximum staging retention is one hour pending G1; no raw media served as static assets.
6. Run fixture-based load/backpressure tests instead of billable stress, plus selected Android/browser recorder/playback, keyboard/screen-reader, zoom/contrast and interruption checks.

## Desired result

The build has reproducible verification evidence and no unresolved release-blocking security, data-loss or truthfulness defect. Device accessibility is checked manually as well as automatically; a green scan alone is not sufficient.

## Concrete outputs and target paths

- .github/workflows/ci.yml, tests/security/, tests/e2e/ and implementation test results.
- apps/api/src/observability/ and apps/api/src/media/cleanup.ts.
- Device/browser/accessibility matrix, fixture-load report and migration/rollback rehearsal evidence.
- Residual risk register with explicit pass/fail/blocked/not-applicable status and owner.

## Exit evidence and rollback

Run all applicable scripts listed in the QA matrix and record exact revision/environment/results. A blocked critical test is not a pass or tacit waiver. If a blocker remains, stop release and disable/revert the smallest faulty feature while preserving revisions/audit. No destructive reset or real-data debugging is permitted. T10 uses only the checked release candidate and documented limitations.

# Phase 10. Prepare the demonstration and complete handoff

Task mapping: T10. Accountable role: Athena; both builders execute. Estimate: 1-2 builder-days. Dependencies: T09. Current status: not started.

Threat coverage: S-04 advice claims, S-11 misleading evidence and S-12 unauthorized publication/scope. Evidence format: claim-to-artifact ledger, rehearsal record and actual human G3 decision.

## Purpose and inputs

Show the real verified workflow in a clear, reproducible demonstration. Inputs are the qualified release candidate, actual synthetic evaluation artifacts, accepted risk register and latest setup instructions. This phase prepares materials; it does not itself authorize public posting.

## Work distribution and actions

1. Freeze the demonstrated build/schema/prompt/model and dataset versions. Rehearse from clean synthetic seed and document exact reset/start steps.
2. Prepare a 60-second story: invented beneficiary selection, short voice input, visible correction/confirmation, supervisor action and honest measured evidence.
3. Include a correction/failure moment rather than suggesting the model never errs. Always show synthetic-data and online-only limitations.
4. State which Gnani APIs/models were actually used and which were only roadmap items. Do not imply telephone/offline/GPU functionality exists.
5. Produce contribution notes for the two actual builders, known limitations, setup/debugging guide and evidence index.
6. Review narration/screens for unsupported accuracy, latency, time-saving or clinical claims; obtain G3 release acceptance before any external action.

## Desired result

A reviewer can understand the user problem, watch the verified end-to-end documentation loop and reproduce the evidence. The handoff makes expected limitations as visible as successes.

## Concrete outputs and target paths

- docs/demo/script.md and docs/demo/evidence.md, with planned/actual claim ledger.
- Recorded demonstration file and associated build/run metadata after the human-approved recording process.
- README.md and SETUP_AND_INITIALIZATION.md updated to actual tested behavior.
- Contribution record, known-issues list and G3 review packet.

## Exit evidence and rollback

Replay the script from the documented starting state, verify every displayed metric against its artifact and confirm no real identity/audio/secret appears. G3 is an actual human decision, not a council signature. If a claim is unsupported, remove or correct it before publication; do not simulate success. External publication remains Phase 11 with explicit authorization.

# Phase 11. Authorize release, publication and submission

Task mapping: T11. Accountable owner: human release owner. Estimate: 0.5-1 builder-day. Dependencies: T10, G3 and explicit deployment/spend/post/submission instructions. Current status: not started.

Threat coverage: S-02 secret exposure, S-08 ongoing spending, S-09 retention and S-12 authorization. Evidence format: approved action record, actual post/submission receipts and release manifest.

## Purpose and inputs

Deliver the approved demonstration through the intended channels without exceeding authorization or the approved cost/privacy boundary. Inputs are the checked release packet, current contest terms, selected hosting/access controls and human instructions.

## Work distribution and actions

1. Recheck official eligibility, deadline, synthetic-data restrictions, Gnani-use and public-post/submission requirements before acting. The source cites November 10, 2026, 23:59 IST; current terms prevail.
2. Obtain explicit authorization for deployment, region/access, spend and the exact public demo/post/submission. A planning request is not permission to communicate publicly on the user's behalf.
3. If remotely deployed, enforce real identity controls and disable fixture identity injection. Use only invented product data, restricted provider usage and safe public error/log behavior.
4. Publish only the approved material through human-authorized actions. No bought engagement, automated vote manipulation or unapproved external messaging.
5. Verify nominated public links and actual submission receipt. Record each member's real contribution and the models/APIs genuinely used.
6. Store a release evidence manifest and monitor the permitted live service within approved cost/retention constraints. Stop or disable unverified paid operations on quota/security failure.

## Desired result

There is a verifiable, authorized release/submission record, not merely a prepared draft or optimistic status. Operational ownership and costs remain clear after the video is made.

## Concrete outputs and target paths

- docs/demo/submission-checklist.md and DECISION_LOG.md release entries.
- Authorized deployment reference/configuration evidence, if deployment is actually approved.
- Nominated public post links and submission receipt, if those actions are approved and completed.
- Release manifest with build/run versions, approver, date, known limitations and rollback owner.

## Exit evidence and rollback

Verify actual links/receipts and the deployed behavior without exposing credentials. A prepared post is not a published post; a submitted form draft is not a receipt. Rollback follows the authorized release plan, preserves audit/data and avoids unapproved account/production changes. A future real-world pilot begins a new approval/design track rather than inheriting this demo's permission.

# 6. Engineering reference: data and interfaces

## 6.0 Exact target-path register

The following paths preserve the roadmap's implementation scope. They are planned files/directories, not existing code. Longer lists wrap for readability; each semicolon separates a distinct path.

| Task | Exact primary target paths |
|---|---|
| T00 | PRD.md; DECISION_LOG.md; DESIGN.md |
| T01 | package.json; package-lock.json; tsconfig.base.json; apps/web/package.json; apps/api/package.json; packages/contracts/package.json; packages/db/package.json; .env.example; .gitignore |
| T02 | apps/api/src/providers/gnani.ts; apps/api/src/media/; tests/contract/; docs/source/VERIFICATION.md |
| T03 | packages/contracts/src/visit.ts; packages/db/prisma/schema.prisma; packages/db/prisma/migrations/0001_initial/migration.sql; packages/db/prisma/seed.ts; apps/api/src/auth/; apps/api/src/policy/assignments.ts |
| T04 | apps/web/src/routes/SignIn.tsx; apps/web/src/routes/NewVisit.tsx; apps/web/src/routes/Visit.tsx; apps/web/src/router.tsx |
| T05 | apps/api/src/extraction/; apps/api/src/routes/extractions.ts; tests/unit/extraction.test.ts |
| T06 | apps/api/src/domain/visits.ts; apps/api/src/routes/visits.ts; tests/integration/visits.test.ts |
| T07 | apps/web/src/routes/ReviewQueue.tsx; apps/web/src/routes/ReviewVisit.tsx; apps/api/src/routes/reviews.ts; tests/e2e/review.spec.ts |
| T08 | eval/fixtures/manifest.json; eval/src/scorer.ts; eval/src/run.ts; eval/tests/scorer.test.ts; apps/web/src/routes/Evaluation.tsx; docs/evaluation/ |
| T09 | .github/workflows/ci.yml; tests/security/; tests/e2e/; apps/api/src/observability/; apps/api/src/media/cleanup.ts |
| T10 | docs/demo/script.md; docs/demo/evidence.md; README.md; SETUP_AND_INITIALIZATION.md |
| T11 | docs/demo/submission-checklist.md; DECISION_LOG.md |

## 6.1 Data model responsibilities

| Entity group | Purpose and required invariant |
|---|---|
| Actor, Organization | Verified issuer/subject maps to active server policy; no client-owned authoritative role |
| Beneficiary, Assignment | Invented references with worker/supervisor scope and organization-consistent foreign keys |
| Visit, VisitRevision | Current state pointer with immutable content revisions; editing clears confirmation |

| Entity group | Purpose and required invariant |
|---|---|
| TransitionAudit | Attributable append-only state/change record; no secret/raw-audio logging |
| IdempotencyRecord | Scoped unique key, request hash and original durable response; changed-body replay conflicts |
| EvaluationRun, EvaluationResult | Synthetic provenance and results isolated from visit mutation privileges |

Proposed database types are UUID identifiers, integer revisions, UTC timestamptz event times, explicit visit timezone, date for confirmed follow-up date, and JSONB for validated observations/provenance. Domain reviewers approve field meanings; engineers do not invent normal/abnormal thresholds.

Initial observation design covers BP systolic/diastolic with mmHg, hemoglobin with g/dL, reported TT-dose mention, optional explicitly confirmed follow-up date and bounded note. A TT mention is documentation of what was said, not a generated vaccination recommendation. Recorded values carry transcript span/manual provenance; unknown/not-measured/not-mentioned values do not silently become clinical defaults.

## 6.2 API distribution and authority

| Contract family | Planned operations and verification |
|---|---|
| Session and roster | GET /session; GET /beneficiaries; verified actor and assigned references only |
| Identity/history | POST /beneficiaries/:id/context; GET /beneficiaries/:id/history; confirmed context plus fresh scope checks |
| Draft and proposal | POST /visits; POST /visits/:id/extractions; PATCH /visits/:id/draft; typed input and expected revision |
| Confirm/submit | POST /visits/:id/confirm; POST /visits/:id/submit; exact snapshot and atomic idempotency |
| Review/correction | GET /reviews; POST /visits/:id/reviews; POST /visits/:id/reopen; assigned supervisor/version and fresh worker confirmation |
| Read-back/results | POST /visits/:id/readback; GET /evaluation/runs/:runId; deterministic confirmed content/read-only synthetic report |

The application prefix is /api/v1. Edit/confirm/submit/review require version preconditions; create/submit/review require idempotency keys as specified in API.md. Use stable validation, identity, stale-conflict, size/media, rate and provider error classes without leaking other beneficiaries or secrets. Endpoint specifications are future contracts, not existing services.

## 6.3 Initial migration and rollback

Review the generated initial SQL, apply to an empty disposable local database, repeat deterministic seed and test constraints/real concurrent transitions. There is no existing dataset to migrate. Future additive changes preserve compatibility; destructive migration needs specific authorization and tested backup/restore. App rollback must not erase submitted revisions or audit history.

## 6.4 Configuration and integration ownership

Server-only planned variable names include DATABASE_URL, GNANI_API_KEY, GNANI_LANGUAGE_CODE, EXTRACTION_MODEL, EXTRACTION_API_KEY, OIDC_ISSUER, OIDC_CLIENT_ID, OIDC_CLIENT_SECRET, SESSION_SECRET, APP_ORIGIN, PROVIDER_MODE and AUDIO_STAGING_DIR. The public frontend may have VITE_API_BASE_URL only as a URL, never a secret. Names are a contract proposal; no values belong in this PDF, git, logs or screenshots.

Oidc vendor, hosting/region, exact versions and retention are unresolved G1 choices. Fixture provider mode aids deterministic development but is conspicuously labeled. It is never an undisclosed substitute for measured Gnani integration or verified identity in a remotely reachable service.

# 7. Validation and measurement strategy

## 7.1 Verification matrix

| Check family | Required evidence | Phase |
|---|---|---|
| Clean install/type/build | Reproducible lockfile, compile and boundary smoke | 01, 09 |
| Unit/schema/semantic | Missing/negative/unit/date/history and confirmation invariants | 03, 05, 06 |
| Provider contracts | Media/header/response shape, timeout/quota/auth, safe logs | 02, 05 |
| DB integration/migration | Scoped queries, immutable snapshots, idempotency/races, empty DB/seed | 03, 06, 07, 09 |
| End-to-end journeys | Worker to return/correct/approve; failures and stale actions | 04, 07, 09 |
| Security | Role spoofing, object access, CSRF/XSS, injection, media/resource abuse | 03, 05, 09 |
| Accessibility/device | Keyboard, screen reader, focus, text, contrast/zoom and actual recorder | 02, 04, 09 |
| Synthetic evaluation | Fixed splits/seeds/hashes, scorer self-tests, reported denominators | 08 |
| Load/operations | Fixture load, bounded concurrency, cleanup/crash recovery and readiness | 09 |
| Demo/release | Claim-to-artifact checks, public authorization and actual receipt | 10, 11 |

## 7.2 Proposed commands, not executed application tests

After T01 creates the scripts, the expected command surface is npm ci; npm run typecheck; npm test; npm run build; npm run test:contract; npm run test:integration; npm run test:e2e; npm run test:security; npm run test:a11y; npm run eval:self-test; npm run test:load:fixture; npm run db:migrate:test; npm run db:seed:test. Command names are a planned interface, not evidence that a test exists or passes today.

Record command, exit status, build revision, fixture/provider mode and environment with each result. A test blocked by missing access or tooling remains blocked. Only include not-applicable cases with a reason and reviewer. No automated scan is a substitute for device accessibility or qualified language/field review.

## 7.3 Metric definitions

Field exact accuracy uses correct structured predictions divided by expected fields under the declared protocol; abstention/missing and error classes remain visible alongside it. Coverage states how many expected fields were actually supplied. Wrong-value rate distinguishes an incorrect supplied value from an acknowledged unresolved field. Schema failure is reported separately from semantic correctness. End-to-end failed cases remain in denominators; conditional metrics may also be shown only with their explicit denominators.

Latency is measured from recording stop to editable draft, including relevant network/provider processing; publish p50/p95 and sample count. Optional TTS first-audio timing is separate. TTS-generated input tests synthetic robustness/self-consistency, not natural ASHA speech. No clinical suitability or real-world time-saving claim follows

from this corpus.

## 7.4 Worked scoring example - illustrative only

Suppose an invented run has 10 expected hemoglobin fields: 7 correct, 1 wrong, 1 abstained and 1 provider failure. Exact end-to-end accuracy is 7/10 = 70%; supplied-field coverage is 8/10 = 80%; wrong-value count is 1/10 expected fields. The provider failure is not removed to make 7/9 appear to be overall success. These numbers illustrate the formula only; they are not SakhiLine test results.

# 8. Security, operations and recovery

## 8.1 Threat-to-control map

Threat IDs match AUTHENTICATION_AND_SECURITY.md: S-01 cross-actor disclosure; S-02 secret theft; S-03 prompt injection; S-04 medical advice output; S-05 wrong value/identity; S-06 duplicate/stale transitions; S-07 malicious media; S-08 billing abuse; S-09 storage/log leakage; S-10 CSRF/XSS; S-11 evaluation contamination/false claims; S-12 unauthorized release or data-scope growth. Phase coverage names planned mitigations, not proof that threats are already resolved.

| Risk | Required boundary | Acceptance evidence |
|---|---|---|
| Wrong beneficiary or cross-worker disclosure | Confirm reference, verified assignment and scoped queries | Duplicate-name and cross-actor denial tests |
| Unconfirmed/stale/duplicate record | Immutable revisions, version check, exact confirmation and idempotent transaction | Real DB races, retry/body mismatch, late proposal tests |
| Prompt injection or advice-shaped system output | No model tools/writes; fixed templates and deterministic confirmed read-back | Adversarial transcript cannot change authority or speak arbitrary advice |
| Media/CPU/billing abuse | Actual content bounds, private staging, restricted conversion, auth/rate/concurrency/spend caps | Invalid/huge/spoofed media, timeout and fixture-load tests |
| Credentials/raw health-like payload leakage | Server-only secrets, safe sessions, redacted logs and no static raw media | Bundle/log inspection and cleanup checks |
| Misleading result/public claim | Frozen artifacts, explicit synthetic labels, actual denominators and human release review | Claim ledger and reproducible report |

## 8.2 Operating the synthetic demo

Proposed baseline is one backend and one PostgreSQL deployment after approval, with liveness/readiness, request IDs, duration/error metadata and pinned configuration. Do not log audio, transcript, beneficiary identifiers, notes, keys or tokens. Use synthetic fixture IDs and operation/request IDs for support reproduction.

Private audio staging is bounded and deleted on success/failure with startup/periodic cleanup for crashes. One-hour maximum staging retention is provisional and needs G1 approval. Synthetic transcripts and evaluation artifacts are access-scoped and retained under the approved policy; no real patient-data retention policy is established by this demonstration.

Do not promise autoscaling, clinical availability, multi-region resilience or production disaster recovery before those designs/tests exist. A service account and evaluator must have minimum necessary privileges. There is no public endpoint for triggering unbounded paid evaluation.

## 8.3 Incident and rollback rules

On provider outage/quota, preserve edits and offer manual/retry within approved policy. On wrong-scope disclosure, real data entered or arbitrary advice-shaped system output, block release and seek human/security review. On data race, disable the faulty mutation path rather than auto-approving or deleting records. Roll back the smallest code change with compatible schema/audit preserved.

Fixture mode is available for controlled local reproduction, not a way to misrepresent live success. Credential rotation, destructive cleanup, production changes or public withdrawal actions need their applicable authorization.

Never print sensitive values while diagnosing a failure.

# 9. Delivery package and future evolution

## 9.1 End-of-project outputs

| Output package | Desired contents and completion evidence |
|---|---|
| Application | Worker and supervisor routes, verified speech/manual flow, version-safe domain and deployed reference only if authorized |
| Data | Reviewed schema, migration, synthetic seed and tested assignment/immutability constraints |
| Integration | Pinned actual provider contract, safe configuration names, bounded failure/retry behavior and authorized usage evidence |
| Evaluation | Frozen manifest/corpus provenance, scorer, versioned results, methods/limitations and no invented metrics |
| QA/operations | Applicable executed results, device/accessibility matrix, cleanup/rollback runbook and residual risks |
| Demo/submission | Accurate 60-second demonstration, contribution record, approved public links and actual submission receipt |
| Documentation | The 18 planning/engineering files updated to actual implemented behavior, with obsolete plans clearly marked |

## 9.2 Definition of project completion

The project is complete for this scope only when the authorized synthetic end-to-end loop works, applicable release checks have actual evidence, claims match measurements, required Gnani use is demonstrated, limitations are visible and human-approved release/submission actions have real receipts. A successful planning review or polished PDF alone does not meet that definition.

## 9.3 Next iteration and separate pilot track

Continuous streaming requires verified PCM capture/framing/pacing and measured segment-to-draft behavior. Durable offline capture requires device quota/eviction/privacy/acknowledgement/retry design; storage.persist() is not a guarantee. PSTN requires caller/worker identity, media/provider/cost contracts and recoverable unconfirmed drafts. Reminders and clinical flags need new domain/policy authorization. GPU Evon and more languages need demonstrated benefit/access/budget rather than award speculation.

Any real-world pilot requires organizational approval, qualified privacy/security/domain review, retention/deletion/export policy, encryption/key custody, incident response and separate usability/field validation. This synthetic contest plan does not establish legal, clinical or production readiness.

# 10. Decision and validation record

## 10.1 Existing council decision

Athena, Draco and Helios each explicitly approved every P1-P6 recommendation in frozen PROPOSAL_R2.md. The exact R2 SHA-256 is ab0ef019095c8dc6fd5147d7369df5b53208b9b83b29beabd1645caf08e9df0b. Final R3 checks retained those votes with no material contradiction. The raw reports disclose previous-context exposure; this was not a fully blind clean-room review.

This PDF is an explanatory delivery plan based on that decision set. Human G0-G3 remains unchanged. Review of this PDF covers fidelity/completeness of the plan, not executable implementation, clinical correctness, actual vendor access or permission to spend.

## 10.2 Human review checklist

- Confirm synthetic contest scope versus a separately designed pilot.
- Confirm language, fluent reviewer, actual two-builder availability and deadline.
- Accept or reject the recording-first reduction from the source live-flow requirement.
- Approve field/unit semantics, identity/hosting, retention and numerical provider spend/retry ceiling.
- Explicitly authorize implementation G2 only after the required prerequisites are resolved.
- Keep deployment, ongoing spend, publication and submission behind G3 and specific action authorization.

## 10.3 Evidence retained with this PDF

The companion review record contains actual Athena product and Draco technical responses bound to the manuscript revision, Helios mechanical/render checks and document hashes. No reviewer is credited merely because a request was sent or a status reported idle. If a member is unavailable or a blocker remains unresolved, the final status must say incomplete rather than imply unanimous validation.

Actual work completed during this authoring task is limited to documentation/PDF generation and its review. No application unit/integration/E2E/security/load/migration or clinical test is claimed to have run. Desired phase results above remain future work.

# 11. Sources, companion documents and glossary

## 11.1 Source and authority order

Human instructions and explicit approvals govern authorization. Frozen council R2 and DECISION_LOG.md govern the recommended scope; implementation roadmap and design contracts elaborate it. The original project brief supplies the larger vision. Public vendor/contest documentation is dated reference material and must be rechecked during integration/release.

| Source | Role in this plan |
|---|---|
| SakhiLine - Project Brief & Architecture.pdf, supplied original, 9 pages | Product vision, channels, stack proposal, contest/deadline assumptions |
| docs/council/PROPOSAL_R2.md and DECISION_LOG.md | Exact recommended P1-P6 scope, actual individual votes and human gates |
| IMPLEMENTATION_ROADMAP.md | T00-T11 ownership, effort ranges, dependencies, acceptance and rollback |
| PRD.md and DESIGN.md | Worker/supervisor journeys, measurable targets and accessible states |
| ARCHITECTURE.md, DATABASE.md and API.md | System boundaries, logical schema, transactional lifecycle and API contracts |
| AUTHENTICATION_AND_SECURITY.md and INTEGRATIONS.md | Threat model, verified-identity requirement and gated provider contracts |
| TESTING_AND_QA.md and DEBUGGING.md | Proposed verification matrix, evidence distinctions and recovery |
| docs/source/VERIFICATION.md | Public-source retrieval and exact limitations; checked 19 October 2026 |

Public references: https://www.gnani.ai/internship ; https://docs.gnani.ai/api/STT/speech-to-text ; https://docs.gnani.ai/api/STT/stt-websocket ; https://docs.gnani.ai/api/TTS/tts-websocket . No current account entitlement, prices or performance is inferred from these pages.

## 11.2 Navigation through the 18 engineering documents

README.md is the starting index. PROJECT_CONTEXT.md is the factual baseline. BACKEND_CONTEXT.md explains proposed backend boundaries. PRD.md, DESIGN.md, ARCHITECTURE.md, DATABASE.md and API.md are product/data/interface designs. AGENTS.md records repository rules. IMPLEMENTATION_ROADMAP.md sequences work. SETUP_AND_INITIALIZATION.md and FEATURE_DEVELOPMENT.md define future setup/change flow. AUTHENTICATION_AND_SECURITY.md, MCP_CONNECTION.md and INTEGRATIONS.md describe trust/tool/provider boundaries. TESTING_AND_QA.md and DEBUGGING.md describe future verification/recovery. DECISION_LOG.md preserves votes, trade-offs and gates.

## 11.3 Glossary

| Term | Meaning here |
|---|---|
| ASHA / supervisor | Worker documentation persona / assigned record reviewer |
| STT / TTS | Speech-to-text / text-to-speech |
| Gnani Prisma / Prisma ORM | Speech API / separate PostgreSQL access tooling |
| Draft / confirmed revision | Editable versioned proposal / exact snapshot explicitly checked by worker |
| Idempotency | Retrying one logical operation returns its original result rather than duplicating it |
| Optimistic version check | Update succeeds only for the expected current revision; stale requests conflict |
| Fixture / live | Deterministic labeled test response / actual authorized provider integration |
| Builder-day | One person's workday estimate, not elapsed calendar time or a delivery guarantee |
| G0-G3 | Human scope/choices, prerequisite decisions, implementation authorization and release/action approval gates |
| Synthetic robustness | Controlled generated-speech performance; not natural-worker or clinical validity |

Final reminder: this is a detailed plan for a synthetic documentation demonstration. The next implementation task is T01 only after the human decision and authorization gates pass.